

CS212: Computer Networks

Introduction to Computer Networks

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Acknowledgement: Many of the slides have been borrowed/revised from various sources including the textbook slides provided by the authors of Computer Networking: A Top-Down Approach and the reference book Computer Networks by Andrew Tanenbaum.

Out line

- Why Computer Networks?
- What is a Computer Network?
- Network Types
- Network Topology
- Introduction to Switching
- Network Performance

Why Computer Networks?

A customary slide, not really needed, but still -- Traditions should be followed whenever it could be followed

- Connectedness – as humans, we inherently long for it
 - Facebook, email, Instagram, ...
- Share information
 - Email, web, ...
- Share resources and provide services
 - Printing, Cloud/Edge computing, ...
- Scalability, Productivity, and Efficiency
 - Great achievements are rarely the work of one

Technological Dominance in History

- 18th Century – Mechanical Systems
- 19th Century – Steam Engine
- 20th Century – Information Technology
 - Telephones, Radio, Television, Computers, Internet
- 21st Century

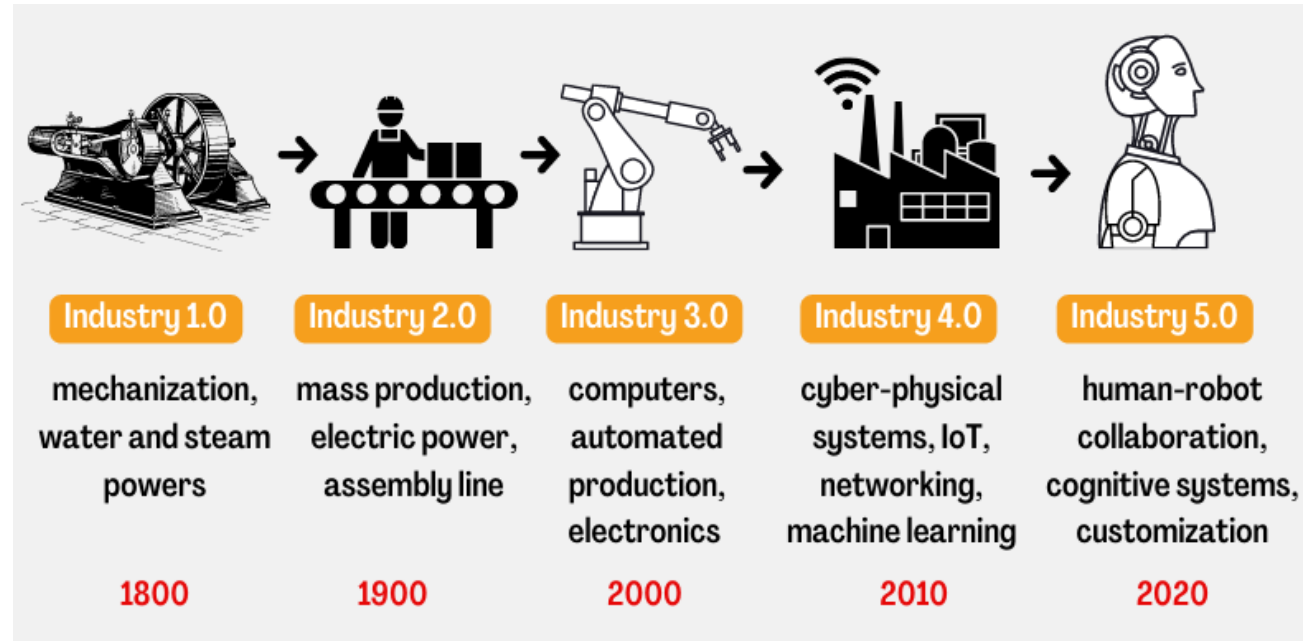


Image source: Internet

- Convergence of Technologies – Evolution of Computing from centralized to distributed computing
- Computer networks is the backbone of modern day computing
- Humans and smart machines collaborate to create sustainable, resilient, and human-centered industries

What is a Computer Network?

1 Computer networking refers to interconnected computing devices that can exchange data and share resources with each other. These networked

2 Computer networking refers to connected computing devices (such as laptops, desktops, servers, smartphones, and tablets) and an ever-expanding array of IoT devices (such as cameras, door locks, doorbells, refrigerators, audio/visual systems, thermostats, and various sensors) that communicate with one another.

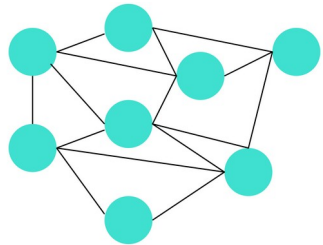
3 Networking, or computer networking, is the process of connecting two or more computing devices, such as desktop computers, mobile devices, routers or applications, to enable the transmission and exchange of information and resources.

4 The **Internet**, a loosely-organized international collaboration of autonomous, interconnected networks, supports host-to-host communication through voluntary adherence to open protocols and procedures defined by Internet Standards.

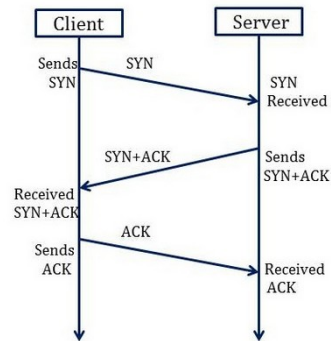
1. <https://aws.amazon.com/what-is/computer-networking/>
2. https://www.cisco.com/c/en_in/solutions/enterprise-networks/what-is-computer-networking.html
3. <https://www.ibm.com/topics/networking>
4. <https://ubiquity.acm.org/article.cfm?id=1071915>

What is a Computer Network?

- A system of **nodes** interconnected via **links** in order to move **information** between nodes and share **resources**
- **Purpose:** Communication between Applications running on different computers
 - Examples Applications: Web browser, Skype, Whatsapp, Multiplayer Games, File sharing ...



Nodes - data terminal equipment (DTE - such as two or more computers and printers) or **data communication equipment** (DCE - such as a switch)



Links - **transmission media** connecting two nodes - may be **physical** (like cable wires or optical fibers), or **free space** used by wireless networks

Protocols - set of rules

*Protocols define the **format**, **order** of **messages sent and received** among network entities, and **actions taken** on message transmission, receipt*

Network Types by Geographical Area

- Personal area network (**PAN**)
 - Serves one (or a very few) person
 - Bluetooth, RFID (Radio Frequency IDentification)
- Local area network (**LAN**)
- Metropolitan area network (**MAN**)
 - Covers a city
 - Example include cable television networks
- Wide area network (**WAN**)
 - Spans a large geographical area, often a country or continent
 - Google, Amazon, Cloudflare, AT&T, Verizon, ...
 - **Internet**

Interprocessor distance	Processors located in same	Example
1 m	Square meter	Personal area network
10 m	Room	
100 m	Building	Local area network
1 km	Campus	
10 km	City	Metropolitan area network
100 km	Country	Wide area network
1000 km	Continent	
10,000 km	Planet	The Internet

Figure 1-6. Classification of interconnected processors by scale.

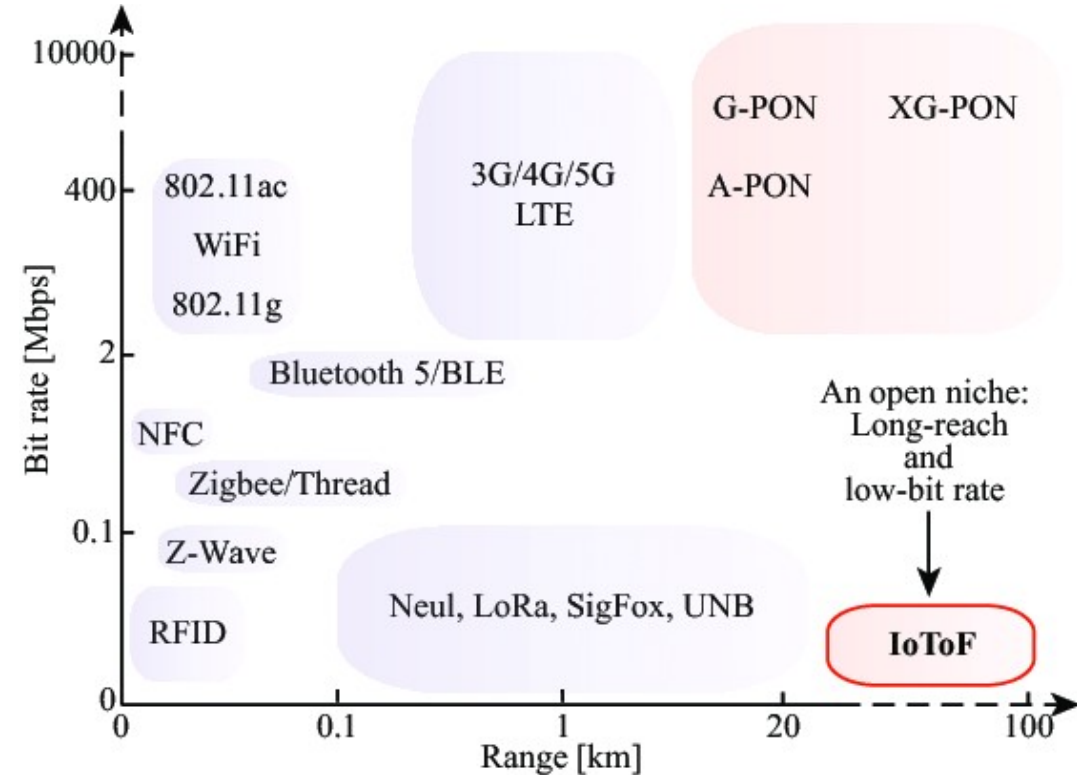
Network Types by Transmission Medium

- **Wired networks**

- Devices are connected by physical wires and cables
- Examples - copper wires and Ethernet, twisted pair, coaxial or fiber optic cables
- Network size and speed requirements determines
 - Choice of cable, topology and the physical distance between devices

- **Wireless networks**

- Infrared, radio or electromagnetic wave transmission across wireless devices with built-in antennae and sensors
- Bluetooth, Cellular networks, WiFi, Satellite communication networks, ...
 - Deep Space Network (DSN)^{1,2}
 - Starlink^{3,4}



1. <https://pam2022.nl/slides/keynotell-Deutsch.pdf>
2. <https://www.youtube.com/watch?v=6WFj-CKldv4>
3. <https://www.starlink.com/ca/technology>
4. <https://www.youtube.com/watch?v=qs2QcycggWU>

Network Types by Communication Type

- Point-to-point networks
 - Direct node-to-node link
- Multipoint networks
 - Multiple devices share channel capacity and network links
- Broadcast networks
 - Several devices receive one-way transmissions from a single sending device
 - Examples include Television stations
- Virtual private networks (VPNs)
 - Secure, point-to-point connection between two network endpoints
 - It establishes an encrypted channel that keeps a user's identity and access credentials, as well as any data transferred, inaccessible to hackers

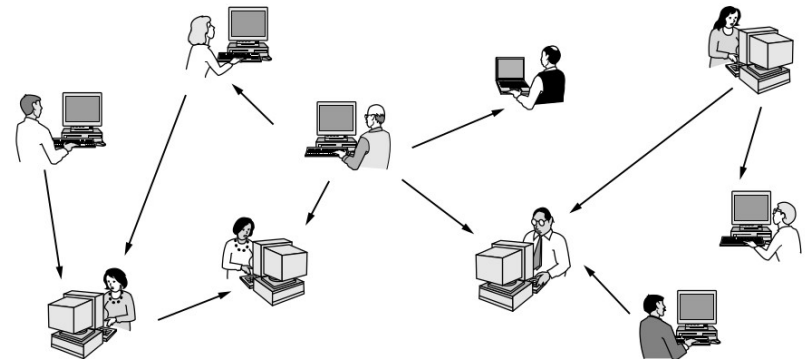
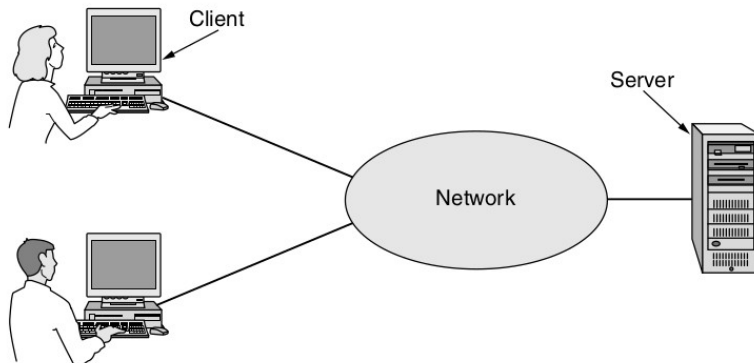
Network Types by Organizational Scope

- Intranet
 - Private network, accessible within an organization
 - AMS, ...
- Internet
 - Network of networks
- Darknet
 - Majority of websites (>90%) is part of the deep web ("deep web" refers to all web pages that are not indexed by search engines, including intranet -- Not accessible to general public)
 - Part of deep web – with illegal/nefarious/anonymous contents
 - Host encrypted websites for its users
 - Decentralized peer-to-peer anonymous networks
 - Need special softwares to access
 - TOR, ...

https://youtu.be/N3-zrhoBx6w?si=wVwy0g02KM4F_P3s

Network Architectures

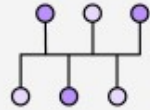
- Client-server architectures
 - (One of more) server(s) – manages resources and delivers services to client devices on the network
 - Clients – don't share their resources and only interact through the server
- Peer-to-peer (P2P) architectures
 - Two or more devices are connected as peers – they have equal power and privileges on the network
 - No central server for coordination
 - Each computer on the network acts as both a client (a computer that needs to access a service) and a server (a computer that provides services to clients)
 - Every peer on the network makes some of its resources available to other network devices, sharing storage, memory, bandwidth and processing power across the network



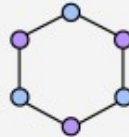
Network Topology

- Network topology describes the
 - **physical** arrangement of nodes and links on a network, including all hardware (routers, switches, cables)
 - **logical** flow of data (irrespective of physical arrangements)

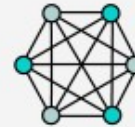
Bus



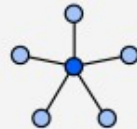
Ring



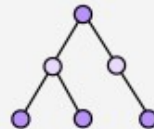
Mesh



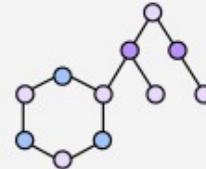
Star



Tree



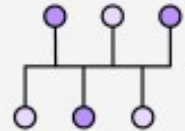
Hybrid



Bus Topology

- All nodes are connected to a single cable—known as the **bus** or **backbone**
 - Like a bus travelling on a bus route
- Data broadcasted to all nodes
 - How to avoid collisions?
- Bus terminator placed at the end of the backbone
 - prevent signal reflection
- Commonly used in small to medium-sized LANs
 - to connect devices such as computers, printers, and other peripherals
 - in industrial control systems and other specialized applications
- Pros -- cost-effective and easy to implement
- Cons
 - Single point of failure (if the backbone fails, the entire network goes down)
 - Less secure (shared backbone)
 - Scalability - with increasing number of nodes the risk of data collisions increases, reducing network efficiency and potentially causing network slowdowns

Bus



Ring Topology

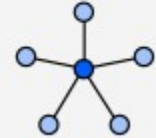
- Nodes are connected in a **circular** fashion, with each node having exactly two neighbors
- Data flows in **one direction** around the ring
- The destination node removes the data from the ring
- **Dual-ring systems** (two concentric rings)
 - can send data in both directions
 - second ring is used when there is a failure in the first ring
- Cheap to install and expand
- Vulnerability -- failure of a single node can bring down the whole network
 - Dual-ring networks are used to protect against this type of failure
 - **Redundancy** minimizes downtime and makes sure that data can continue flowing if one ring fails



Star Topology

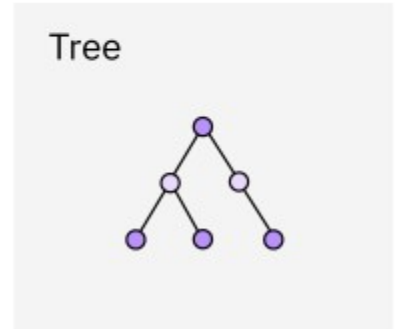
- All nodes are connected to a **central hub/switch**
- If a single node fails, the rest of the network is unaffected, as long as the central hub is operational
- Easy to troubleshoot and manage
 - Popular choice for LANs
- Its centralized structure also makes adding or removing devices relatively easy, contributing to its scalability
- Performance of the entire network depends on the central hub/switch and the connections to it
 - If the central hub/switch goes down, then the entire network goes down with it

Star



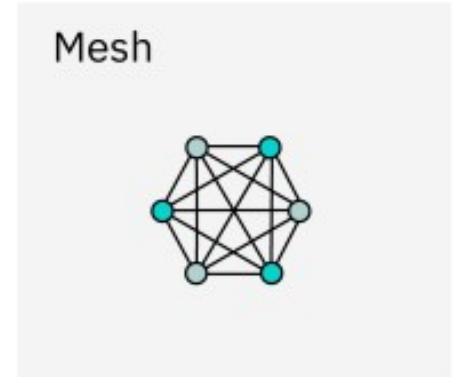
Tree Topology

- A hierarchical structure by combining both bus and star networks
- A central hub serves as the root node, connecting to multiple star networks rather than individual nodes
- Enables a higher number of devices to connect to a central node, enhancing data flow efficiency
- Like star networks, tree topologies facilitate straightforward identification and resolution of issues with individual nodes
- Network nodes depend on a central hub, creating dependencies that can affect network performance
- Tree topologies also inherit vulnerabilities from both bus and star networks
 - Single point of failure -- central hub



Mesh Topology

- A highly interconnected network structure where each node is directly linked to multiple other nodes
- **Full** mesh configuration -- every node connects to every other node within a single network, creating redundant paths for data transmission
 - **Alternative paths** -- enhanced network resilience and fault tolerance
- **Partial** mesh topologies – a node is directly connected to some other nodes
 - Cost-effective
- **Decentralized** structure -- eliminate single point of failure, enhancing both security and efficiency
- Improved data transmission speeds and scalability
- Increased complexity in network design and management
- Common in
 - critical infrastructure,
 - wireless networks
 - scenarios requiring high reliability and performance

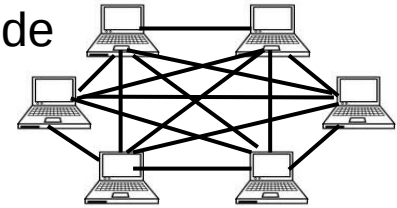


Switching

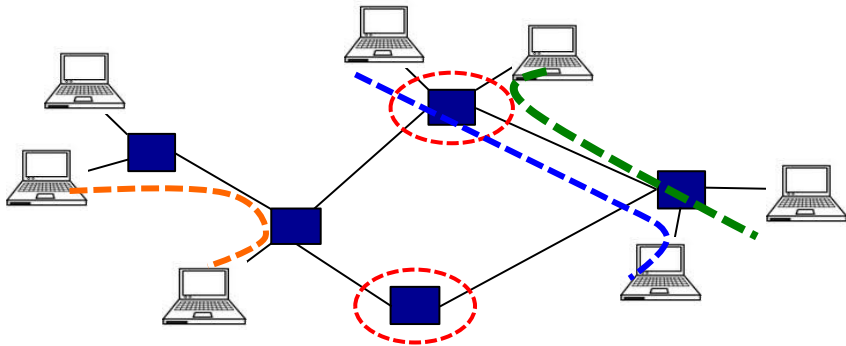
- Recall a network is set of nodes and links
- Connecting two nodes is trivial



- What if we have more nodes?
 - One link for every node
 - Not scalable



- Solution: A switched network – Nodes share network link resources
- Switching can be implemented as packet switching or circuit switching
- Circuit switching – used in the Public Switched Telephone Network (PSTN)
- Packet switching – used in the Internet



Conventional manual telephone switchboards (on right), now replaced by automated (Digital Circuit Switching) Systems.

Note, mobile networks and VoIP services are based on packet switching instead of circuit switching.



We will talk more about circuit switching and (a lot more about) packet switching throughout this course

Network Performance

- What to measure
- One or more of the following may be relevant
 - Latency/delay - processing delay, queuing delay, transmission delay, and propagation delay
 - Throughput - rate of message delivery
 - Utilization
 - Loss
 - Reliability/Resilience